

Simultaneous Equations - Worded Problems

Exercise A

1. 3 pens and 5 rulers cost £7.60
4 pens and 2 rulers cost £6.40
Find the cost of one pen and one ruler.
2. 2 pencils and 5 erasers cost £3.90
3 pencils and 4 erasers cost £4.10
Find the cost of each item.
3. 2 notebooks and 5 rulers cost £8.10
3 notebooks and 4 rulers cost £8.30
Find the cost of each item.
4. 3 burgers and 5 drinks cost £13.60
4 burgers and 3 drinks cost £13.00
Find the cost of a burger and a drink.
5. 2 tickets and 5 snacks cost £16.00
3 tickets and 4 snacks cost £17.00
Find the cost of a ticket and a snack.
6. 3 apples and 5 bananas cost £2.70
4 apples and 2 bananas cost £2.20
Find the cost of each fruit.
7. 3 pens and 5 rulers cost £5.80
4 pens and 2 rulers cost £4.00
Find the cost of each item.
8. 3 shirts and 5 ties cost £50
4 shirts and 3 ties cost £52
Find the cost of a shirt and a tie.

Exercise B

9. 3 bags and 5 hats cost £19
4 bags and 3 hats cost £18
Find the **total** cost of 2 bags and 7 hats.
10. 2 pizzas and 5 sides cost £14.90
3 pizzas and 4 sides cost £13.60
Find the **total** cost of 6 pizzas and 1 side.
11. 3 jumpers and 5 scarves cost £19.50
4 jumpers and 3 scarves cost £20.50
Find the **total** cost of 1 jumper and 6 scarves.
12. 2 games and 5 controllers cost £20
3 games and 4 controllers cost £19.50
Find the **total** cost of 4 games and 1 controller.
13. 3 muffins and 5 cookies cost £7.80
4 muffins and 3 cookies cost £6
Find the **total** cost of 5 muffins and 1 cookie.
14. 3 lamps and 5 bulbs cost £25
4 lamps and 3 bulbs cost £26
Find the **total** cost of 2 lamps and 7 bulbs.
15. 2 milkshakes and 5 biscuits cost £6.10
3 milkshakes and 4 biscuits cost £7.40
Find the **total** cost of 6 milkshakes and 1 biscuit.
16. 5 jackets and 3 caps cost £44
3 jackets and 5 caps cost £36
Find the **total** cost of 2 jackets and 4 caps.

Answers - Exercise A

1. Pen = £1.20 Ruler = £0.80
2. Pencil = £0.70 Eraser = £0.50
3. Notebook = £1.30 Ruler = £1.10
4. Burger = £2.20 Drink = £1.40
5. Ticket = £3.00 Snack = £2.00
6. Apple = £0.40 Banana = £0.30
7. Pen = £0.60 Ruler = £0.80
8. Shirt = £10 Tie = £4

Answers - Exercise B

9. £20
10. £9.70
11. £13
12. £13
13. £4.20
14. £24
15. £11.30
16. £26

Exercise C

17. A greengrocer sells apples and pears at fixed prices.
A customer buys 3 apples and 5 pears for £5.00.
Another customer buys 4 apples and 3 pears for £4.10.
A third customer buys only pears and spends £3.50. How many pears did they buy?
18. A bookshop sells books and pens.
2 books and 5 pens cost £11.50.
3 books and 4 pens cost £12.00.
A student has £10 and wants to buy as many pens as possible.
What is the maximum number of pens they can buy?
19. At a cinema, tickets and snacks are sold.
3 tickets and 5 snacks cost £24.50.
4 tickets and 3 snacks cost £23.50.
A family buys 2 tickets and 2 snacks.
How much change do they receive from £20?
20. A shop sells bags and hats.
3 bags and 5 hats cost £39.
5 bags and 2 hats cost £46.
A customer buys 1 bag and 4 hats.
They pay with £20.
Do they receive any change? Justify your answer.
21. A stationery shop sells notebooks and rulers.
2 notebooks and 5 rulers cost £9.60.
3 notebooks and 4 rulers cost £10.20.
A teacher buys 10 rulers.
How much more would it cost to instead buy 5 notebooks?
22. At a café, juices and sandwiches are sold.
3 juices and 5 sandwiches cost £22.00.
4 juices and 3 sandwiches cost £19.25.
A customer buys 1 juice and 1 sandwich.
Another customer buys 2 juices.
Who spends more, and by how much?

Answers - Exercise A

17. 5 pears
18. 6 pens
19. £7.00 change
20. Total = £20 → no change
21. 10 rulers = £12, 5 notebooks = £9 → notebooks are £3 cheaper
22. Juice + sandwich = £4.00, 2 juices = £2.50 → first customer spends £1.50 more